

TECHNICAL SUMMARY SHEET

Free C-Program API Library for applications to easily and fail-safely generate log messages.

This C-Program Library is free to use under the MIT License agreement. It provides an easy-to-use printf() style API function interface to generate application log messages:

```
LoggerAppsMsg( DeOp, LogicalName, SeverityLevel, "Format string specification, string specifier value %s", "string value" );
```

Technical benefits of this API function:

1. DeOp directs generating log messages during development, production or both.
2. LogicalName indicates file name, function name or common name across multiple files.
3. SeverityLevel indicates different levels of severity importance.
4. The remainder of the arguments are the same as printf() function.
5. API Arguments Validation:
 - a. Allow generating log messages to development, production or both.
 - b. Use only LogicalName defined for the instance of the API.
 - c. Use only SeverityLevel defined for the instance of the API.
 - d. Scan printf() format string specification for string specifiers and that match corresponding arguments.
6. API Bound Argument Validation:
 - a. Validate length of format string specification to prevent unbounded strings.
 - b. Validate length of string argument to prevent improperly NULL terminated strings.
 - c. Validate specific ASCII characters for generating log messages to prevent unwanted ASCII characters in log message.
 - d. Validate against an acceptable set of format string specifiers to prevent unsafe and malicious attacker patterns.
7. Robust and Fail-Safe API Design:
 - a. Application does not need to check return status from the API.
 - b. Most failures should not impact application execution flow.
8. Standard Output Log File Content:

```
02/09/2026 12:01:55 [3650:sample_program] [DATABASE] [INFO] Message that  
generates 1 string and 2 integer arguments [String Argument], [98] and [99]
```

where:

```
"02/09/2026 12:01:55" date and timestamp when the log message was generated.  
"[3650:sample_program]" application process id and name.  
"[DATABASE]" logical message group for related application log messages.  
"[INFO]" severity for log message.
```

```
The remainder of the output is derived from the C-Programming Language printf()  
format string specification and corresponding string specifier data value.
```

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Download: <https://github.com/wandi-ssal/secure-application-logger>